

BMK Tropisetron

Choco MST

Mual & Muntah

Mual / nausea

Muntah / vomiting / emesis

Retching = mau muntah tapi ga keluar (muntah kering)

Kenapa mual-muntah dianggap serius

Dehidrasi → lemas, pusing, BAK sedikit.

Gangguan elektrolit (mineral tubuh kacau) → lemah, kram, bisa ganggu jantung.

Aspirasi (muntah masuk paru) → risiko pneumonia aspirasi

Bagaimana Bisa Muntah?

Tubuh kita punya **Pusat Muntah** di otak
Komando ini menerima sinyal dari beberapa lokasi

Apa saja yang berhubungan dengan MUNTAAH?

Usus & lambung = pemberi
sinyal

Saat ada rangsangan
(kemoterapi, iritasi,
distensi/perut kembung), sel
di usus bisa melepas
serotonin

Saraf vagus = kabel dari
usus ke otak → mengirim
sinyal ke Pusat Muntah Otak

Kalau serotonin menekan
tombol tertentu di ujung
saraf vagus, sinyal mual akan
naik ke otak.

CTZ (Chemoreceptor
Trigger Zone) = sensor
mendeteksi obat/racun di
otak

Kalau ada obat/toksin →
CTZ kerangsang → kirim
sinyal ke pusat muntah

Pusat muntah otak = Jika
ada dapat sinyal →
merangsang muntah

Sumber Sinyal ke Pusat Muntah

Lambung / usus (iritasi, kemo, peregangan) → kirim sinyal lewat saraf vagus
Pengirim sinyal: 5-HT3 (serotonin)

CTZ = “sensor obat/racun” di otak (darah membawa pemicu)
Pengirim sinyal: D2 (dopamin), 5-HT3, NK1

Telinga bagian dalam (vestibular) = mabuk perjalanan/vertigo
Pengirim sinyal: H1 (histamin), M1 (muskarinik)

Pusat nyeri/emosi di otak (cemas, takut, bau menyengat, trauma kemo)

Korteks & batang otak (refleks khusus)
Pengirim sinyal: Batuk hebat, menelan darah, rangsang tenggorokan → refleks muntah

4 Pengirim Sinyal Utama

5-HT₃ (serotonin
type-3)

Banyak di lambung /
usus (ujung saraf
vagus) dan juga di jalur
pusat otak

Sangat dominan pada
CINV akut dan
PONV/RINV

5-HT₃ Agonist

Dopamin (D₂)

Banyak di CTZ.

Domperidone
Metoclopramide
Chlorpromazine

NK₁ (Substance P)

Penting untuk muntah,
terutama CINV (sering
kombinasi regimen
pada risiko tinggi)

Aprepitant
Fosaprepitant
Rolapitant

Histamin (H₁) &
muskarinik (M₁)

Lebih dominan di
mabuk
perjalanan/vestibular

Dimenhydrinate
Diphenhydramine

Apa itu 5-HT₃?

5-HT₃ / Serotonin = tombol yang kalau aktif mengirim sinyal memicu mual & muntah.

Lokasi 5-HT₃ banyak di lambung/usus

Rangsangan



5-HT₃ aktif



Lewat Saraf Vagus



Otak
(Pusat Muntah)



Mual & Muntah

Tipe-Tipe Serotonin Dalam Tubuh

Serotonin bekerja lewat 7 keluarga reseptor: 5-HT₁ samapi 5-HT₇ (totalnya banyak sub-tipe)

5-HT₁:

banyak di otak
Efek:
mood, cemas,
nyeri, juga target
beberapa obat
saraf/migrain

5-HT₂:

banyak efek ke
pembuluh darah
& otot polos
(kontraksi/relaks
asi)

5-HT₃:

Paling terkenal
untuk
mual-muntah
(jalur
usus-otak).

5-HT₄:

Banyak di
saluran cerna →
membantu
gerak
usus/peristaltik
(motilitas).

5-HT₅:

Masih kurang
jelas pada
manusia
dibanding yang
lain

5-HT₆:

Dominan di area
otak terkait
kognisi/memori
(banyak riset ke
arah gangguan
kognitif).

5-HT₇:

Fungsi saraf dan
peran di usus
(relaksasi otot
polos kolon)

Apa Saja Yang Merangsang 5-HT₃

Kemoterapi emetogenik (cisplatin)

Radiasi/iritasi saluran cerna (terutama area yang mengganggu saluran cerna)

Infeksi/toksin di usus (Gastroenteritis)

Distensi / perut penuh banget

Rangsang kimia di saluran pencernaan

Obat-obatan Opioid

Apa itu PONV?

PONV = mual (nausea) dan/atau muntah (vomiting) yang muncul setelah operasi (sering dalam 24 jam pertama)

Termasuk efek samping yang sering dan mengganggu pemulihan
Pasien jadi nggak nyaman, sulit makan/minum → bisa tambah lama rawat

Penyebabnya campuran: obat anestesi, opioid (obat nyeri), jenis operasi (saluran pencernaan)

Seberapa sering PONV terjadi?

**20 –
30%**

20–30% pada populasi bedah
secara umum

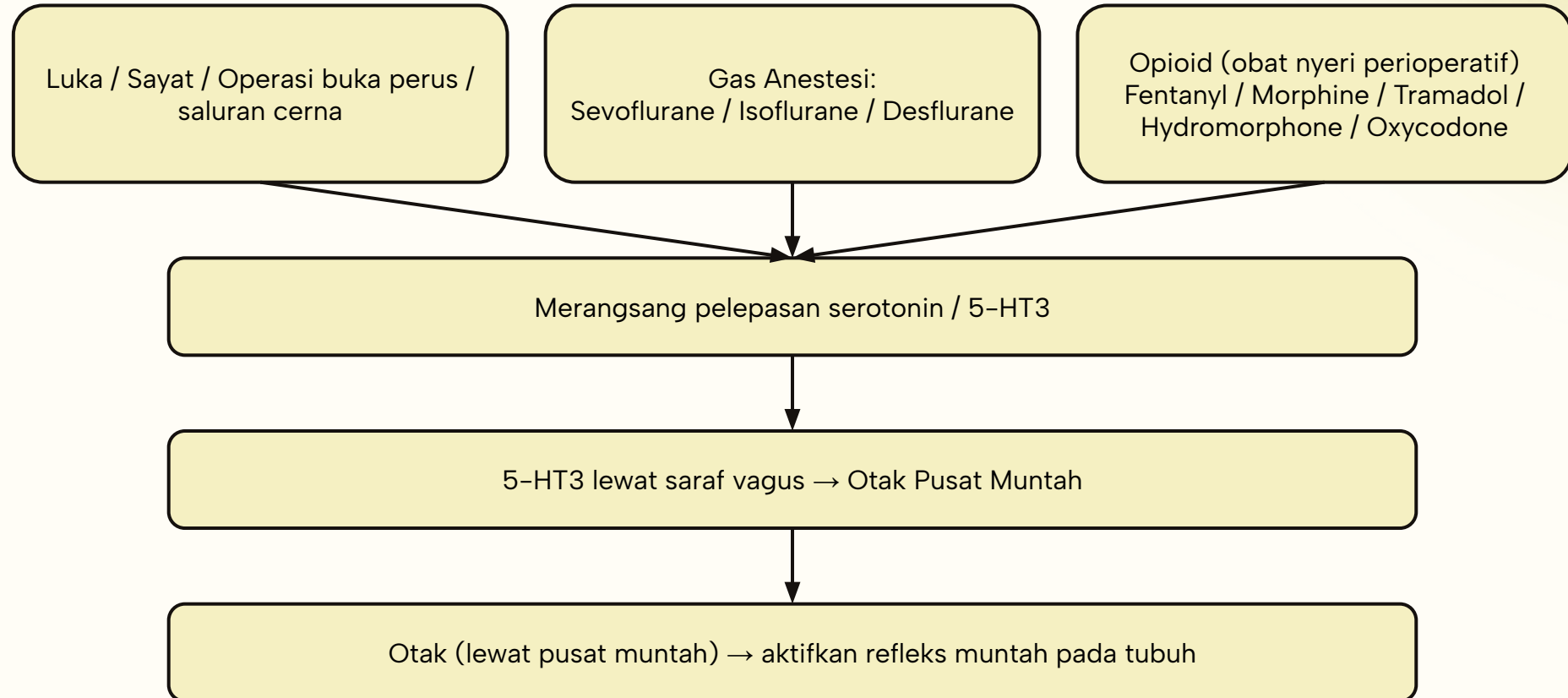
**70 –
80%**

bisa naik jadi 70–80% (mis.
pasien dengan faktor risiko
lebih banyak)

**30 –
80%**

Pada pasien dewasa dengan
anestesi umum bisa 30–80%

Bagaimana Dapat Terjadi PONV?



Faktor Risiko PONV

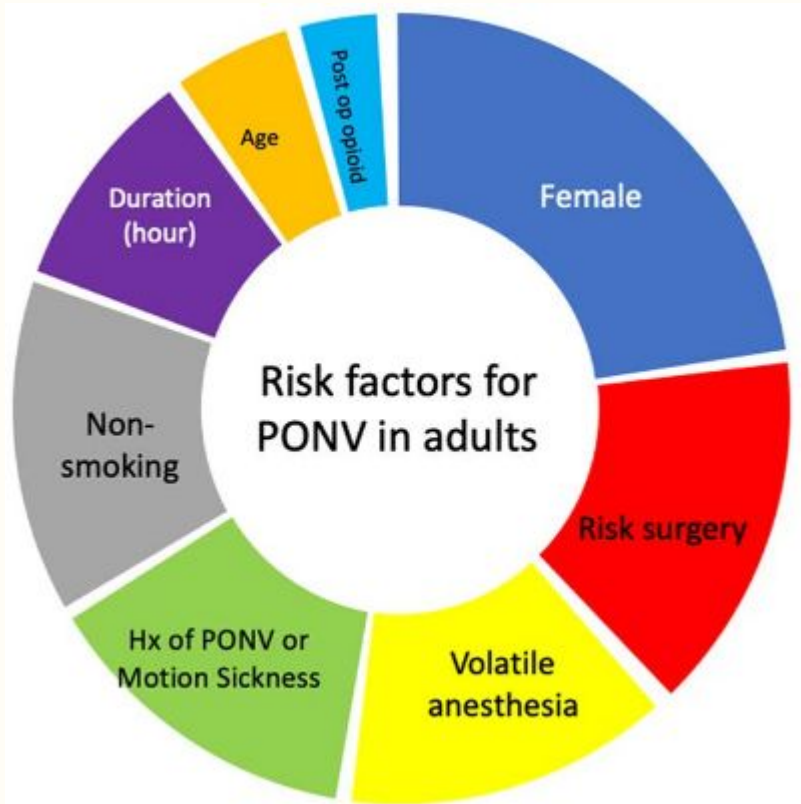


Table 2. Risk Factors for PONV in Adults

Evidence	Risk Factors
Positive overall	Female sex (B1)
	History of PONV or motion sickness (B1)
	Nonsmoking (B1)
	Younger age (B1)
	General versus regional anesthesia (A1)
	Use of volatile anesthetics and nitrous oxide ^a (A1)
	Postoperative opioids (A1)
	Duration of anesthesia (B1)
	Type of surgery (cholecystectomy, laparoscopic, gynecological) (B1)
Conflicting	ASA physical status (B1)
	Menstrual cycle (B1)
	Level of anesthesiologist's experience (B1)
	Perioperative fasting (A2)
Disproven or of limited clinical relevance	BMI (B1)
	Anxiety (B1)
	Nasogastric tube (A1)
	Migraine (B1)
	Supplemental oxygen (A1)

Abbreviations: ASA, American Society of Anesthesiologists; BMI, body mass index; PONV, postoperative nausea and vomiting.

^aUse of nitrous oxide over 1 h duration.

Guideline Penggunaan 5-HT₃ pada PONV

Fourth Consensus Guidelines PONV:
5-HT₃ receptor antagonists adalah salah satu “pilar” profilaksis PONV

and studied 5-HT₃ receptor antagonist and is considered the “gold standard” in PONV management (evidence A1).¹⁹⁹ It has comparable antiemetic and antinausea

Dosis Tropisetron pada PONV

Tropisetron 2 mg IV (timing: end of surgery) → Profilaksis Dewasa

0.1mg/kgBB sampai 2mg → Profilaksis Anak

5 mg IV sebelum anestesi dilaporkan efektif pada bedah payudara & ginekologi

Table 4. Antiemetic Doses and Timing for Prevention of PONV in Adults

Drugs	Dose	Evidence	Timing
Amisulpride	5 mg	A2 ^{113,114}	At induction
Aprepitant	40 mg PO	A1 ¹¹⁵⁻¹¹⁷	At induction
Casopitant	150 mg PO	A1 ¹¹⁹⁻¹²¹	At induction
Dexamethasone	4–8 mg IV	A1 ¹²²	At induction
Dimenhydrinate	1 mg/kg IV	A1 ¹²⁴⁻¹²⁶	
Dolasetron	12.5 mg IV	A2 ¹²⁷⁻¹²⁹	End of surgery; timing may not affect efficacy
Droperidol ^a	0.625 mg IV	A1 ^{130,131}	End of surgery
Ephedrine	0.5 mg/kg IM	A2 ^{133,134}	
Granisetron	0.35–3 mg IV	A1 ^{135,136}	End of surgery
Haloperidol	0.5 to <2 mg IM/IV	A1 ¹⁴⁰	
Methylprednisolone	40 mg IV	A2 ¹⁴¹	
Metoclopramide	10 mg	A1 ¹⁴²	
Ondansetron	4 mg IV	A1 ^{143,144}	End of surgery
	8 mg PO or ODT		
Palonosetron	0.075 mg IV	A1 ¹⁴⁶⁻¹⁴⁸	
Perphenazine	5 mg IV	A1 ¹⁴⁹	
Promethazine ^a	6.25 mg	A2 ^{150,151}	
Ramosetron	0.3 mg IV	A1 ¹⁵²	End of surgery
Rolapitant	70–200 mg PO	A3 ¹⁵³	At induction
Scopolamine	Transdermal patch	A1 ^{154,155}	Prior evening or 2 h before surgery
Tropisetron	2 mg IV	A1 ¹⁵⁷	End of surgery

Table 6. Antiemetic Doses for Prophylaxis of POV/ PONV in Children

Drug	Dose	Evidence
Aprepitant	3 mg kg ⁻¹ up to 125 mg	A3 ³⁵⁶
Dexamethasone	150 µg kg ⁻¹ up to 5 mg	A1 ¹⁹⁶
Dimenhydrinate	0.5 mg kg ⁻¹ up to 25 mg	A1 ¹²⁴
Dolasetron	350 µg kg ⁻¹ up to 12.5 mg	A2 ³⁵⁷
Droperidol ^a	10–15 µg kg ⁻¹ up to 1.25 mg	A1 ¹³²
Granisetron	40 µg kg ⁻¹ up to 0.6 mg	A2 ³⁵⁸
Ondansetron ^b	50–100 µg kg ⁻¹ up to 4 mg	A1 ³⁵⁹
Palonosetron	0.5–1.5 µg kg ⁻¹	A2 ^{360,361}
Tropisetron	0.1 mg kg ⁻¹ up to 2 mg	A1 ¹⁵⁷

Apa itu CINV?

CINV = mual/muntah akibat kemoterapi (obat kanker)

Tipe CINV:

Anticipatory (mual sebelum kemo)
Breakthrough (tetap mual walau sudah dicegah)
Refractory (gagal di siklus sebelumnya & berulang)

Dibagi berdasarkan waktu:
Acute (0–24 jam)
Delayed (>24 jam – berhari2)

Risiko CINV dipengaruhi:

Kekuatan Obat pemicu muntah
HEC tinggi
MEC sedang

Faktor Pasien:
Perempuan
Usia lebih muda
Riwayat CINV sebelumnya

HEC & MEC

HEC

Highly Emetogenic Chemotherapy

Risiko muntah SANGAT tinggi

>90% pasien bisa muntah dalam 24 jam pertama
kalau tanpa obat pencegahan

- Cisplatin
- Carmustine
- Dacarbazine
- Mechlorethamine
- Streptozocin
- Cyclophosphamide $\geq 1500 \text{ mg/m}^2$
- kombinasi anthracycline + cyclophosphamide (AC)

Kanker yang menggunakan HEC:
Testis, paru, kandung kemih, serviks, ovarium,
payudara

MEC

Moderately Emetogenic Chemotherapy

Risiko muntah SEDANG–cukup tinggi

30–90% pasien bisa muntah tanpa pencegahan

- Carboplatin
- Oxaliplatin
- Irinotecan

Kanker yang menggunakan MEC:
Ovarium
NSCLC (paru)
Kolorektal

Faktor Risiko CINV

Riwayat Sebelumnya:

- Mengalami muntah setelah siklus kemoterapi sebelumnya.
- Memiliki riwayat mual dan muntah di masa lalu
- Mengalami mual dan muntah antisipatif (sebelum tindakan dilakukan)

Karakteristik Pasien:

- Berjenis kelamin perempuan
- Berusia kurang dari 60 tahun
- Memiliki riwayat mabuk perjalanan (motion sickness) atau mual terkait kehamilan

Tindakan Medis & Obat-obatan:

- Jenis obat kemoterapi yang diberikan (terutama regimen yang mengandung platinum atau anthracycline)
- Pemberian obat antiemetik (anti mual)
- Sedang menjalani siklus kemoterapi pertama, kedua, atau ketiga

Gaya Hidup & Kebiasaan:

- Konsumsi alkohol
- Tidur kurang dari tujuh jam pada malam sebelum kemoterapi
- Melakukan pengobatan mandiri dengan obat-obatan bebas (over-the-counter)

Seberapa sering CINV terjadi?

55,3%

Acute CINV 55,3%

62,3%

Delayed CINV 62,3%

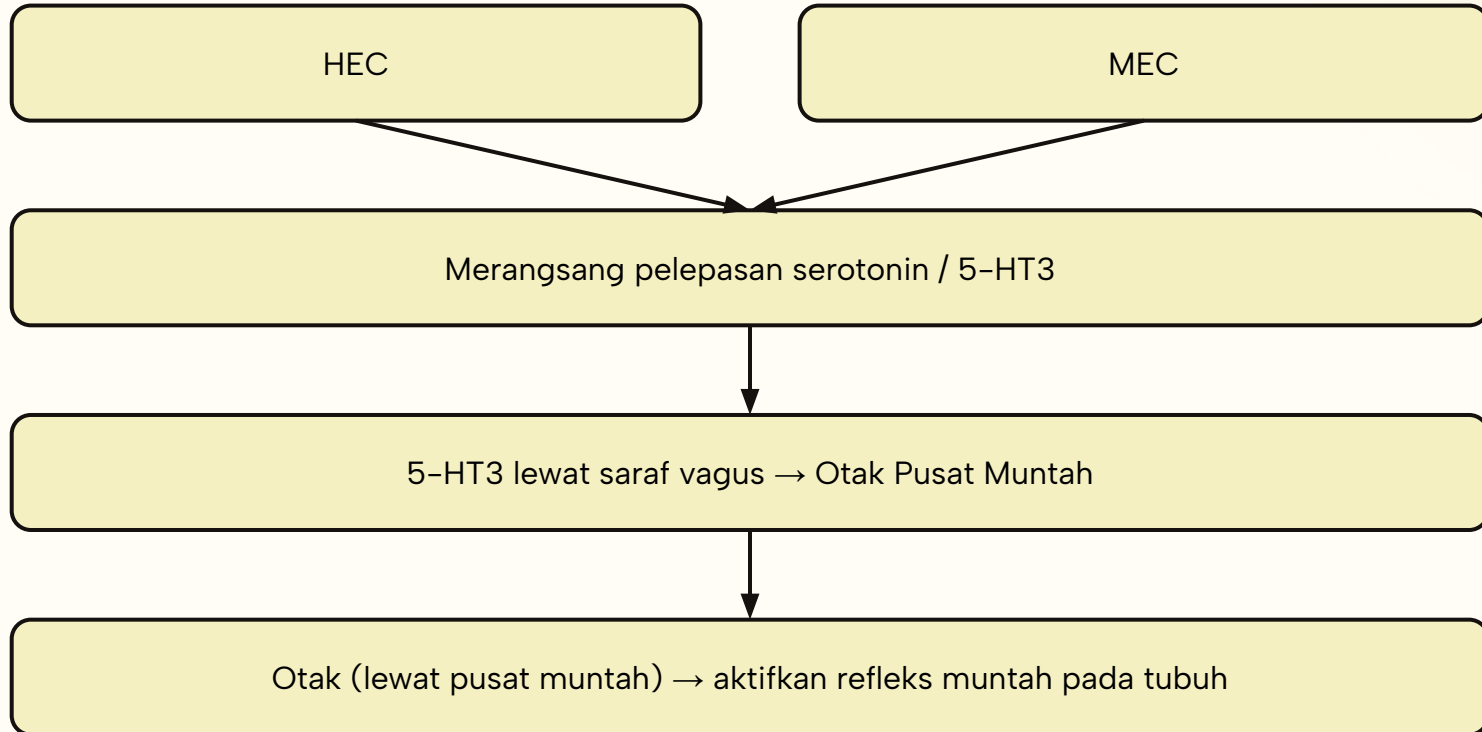
**70 –
80%**

70–80% pasien dewasa kanker
mengalami CINV

>90%

HEC tanpa pencegahan,
muntah dapat terjadi >90%
pasien

Bagaimana Dapat Terjadi CINV?



Penggunaan 5-HT₃ pada CINV

ACUTE Nausea and Vomiting: SUMMARY

EMETIC RISK GROUP	ANTIEMETICS				
High Non-AC	5-HT ₃	+	DEX	+	NK ₁ +/- OLZ*
High AC	5-HT ₃	+	DEX	+	NK ₁ +/- OLZ*
Carboplatin	5-HT ₃	+	DEX	+	NK ₁
Moderate (other than carboplatin)	5-HT ₃	+	DEX		
Low	5-HT ₃	or	DEX	or	DOP
Minimal	No routine prophylaxis				
5-HT ₃ = serotonin ₃ receptor antagonist	DEX = DEXAMETHASONE	NK ₁ = neurokinin ₁ receptor antagonist such as APREPITANT or FOSAPREPITANT or ROLAPITANT or NEPA (combination of netupitant and palonosetron)		OLZ = OLANZAPINE	DOP = dopamine receptor antagonist

NOTE: If the NK₁ receptor antagonist is not available for AC chemotherapy, palonosetron is the preferred 5-HT₃ receptor antagonist.

* OLZ: Olanzapine may be added particularly if nausea is a concern.



Apa itu RINV?

RINV = mual/muntah akibat radioterapi (terapi sinar)

Bisa muncul selama rangkaian radiasi atau setelah sesi radiasi

Risiko beda-beda tergantung lokasi radiasi (mis. kena perut atas lebih sering) dan dosis per sesi

40–80% pasien bisa mengalami mual/muntah

Seberapa sering RINV terjadi?

<30%

payudara, ekstremitas

**30–60
%**

cranium/kepala, craniospinal,
head-neck, pelvis

**60–90
%**

upper abdomen (perut atas)

>90%

TBI (radiasi seluruh tubuh), TNI
(radiasi kelenjar getah bening
seluruh tubuh)

Penggunaan 5-HT3 pada RINV

Table 1 Radiotherapy emetic risk levels and MASCC/ESMO antiemetic guideline update 2023

Emetic risk level	Area of treatment	Antiemetic guideline	Level of evidence/grade of recommendation
High	Total body irradiation	Prophylaxis with a 5-HT ₃ -RA + DEX	II/B (for the addition of DEX: III/C)
Moderate	Upper abdomen, craniospinal	Prophylaxis with a 5-HT ₃ -RA + optional DEX	II/A (for the addition of DEX: II/C)
Low	Brain	Rescue with DEX	IV/B
	Head & neck, thorax, pelvis	Rescue with DEX, a dopamine RA, or a 5-HT ₃ -RA	IV/B
Minimal	Extremities, breast	Rescue with DEX, a dopamine RA, or a 5-HT ₃ -RA	IV/B
Concomitant RT and weekly cisplatin 40 mg/m ²	Acute NV: prophylaxis day 1 before administration of cisplatin with a 5-HT ₃ -RA, DEX, and an NK ₁ -RA Delayed NV: DEX days 2-4		II/B
Concomitant CRT	In concomitant CRT, the antiemetic prophylaxis is according to the CT-related antiemetic guidelines of the corresponding risk category; unless, the risk of emesis is higher with RT than CT		IV/B

Dexamethasone on days 2–4 was administered in the clinical trial including cervical cancer patients (younger women, pelvic radiation). No trial has compared dexamethasone day 1 vs days 2–4, and no trial has explored the optimal regimen in other treatment settings

5-HT₃-RA 5-hydroxytryptamine₃-receptor antagonist, CT chemotherapy, CRT chemoradiotherapy, DEX dexamethasone, NV nausea and vomiting, RT radiotherapy

Penggunaan 5-HT₃ pada RINV

Table 1. Summary of Recommendations (continued)

Clinical Situation	2006 Recommendation	2011 Recommendation
Radiation-induced nausea and vomiting		
High risk	No change from original guideline. The Update Committee suggests administration a 5-HT ₃ antagonist with or without a corticosteroid before each fraction and for at least 24 hours after. There is no change from the original guideline.	On the basis of extrapolation from indirect evidence, the Update Committee recommends that all patients should receive a 5-HT ₃ antagonist before each fraction and for at least 24 hours after completion of radiotherapy. Patients should also receive a 5-day course of dexamethasone during fractions 1-5.
Moderate risk	The Update Committee recommends a 5-HT ₃ antagonist before each fraction.	The Update Committee recommends that patients receive a 5-HT ₃ antagonist before each fraction for the entire course of radiotherapy. Patients may be offered a short course of dexamethasone during fractions 1-5.
Low risk	No change from original guideline. The Update Committee recommends a 5-HT ₃ antagonist before each fraction.	The Update Committee recommends a 5-HT ₃ antagonist alone as either prophylaxis or rescue. For patients who experience radiation-induced nausea and vomiting while receiving rescue therapy only, prophylactic treatment should continue until radiotherapy is complete.
Minimal risk	No change from original guideline. The Update Committee suggests that treatment be administered on an as-needed basis only. Dopamine or serotonin receptor antagonists are advised. Antiemetics should be continued prophylactically for each remaining radiation treatment day.	Patients should receive rescue therapy with either a dopamine receptor antagonist or a 5-HT ₃ antagonist. Prophylactic antiemetics should continue throughout radiation treatment if a patient experiences radiation-induced nausea and vomiting while receiving rescue therapy.
Combined chemotherapy and radiation therapy	Patients should receive antiemetic prophylaxis according to the emetogenicity of chemotherapy, unless the emetic risk with the planned radiotherapy is higher. No change from the original guideline.	Patients should receive antiemetic prophylaxis according to the emetogenicity of chemotherapy, unless the emetic risk with the planned radiotherapy is higher. No change from the original guideline.

Abbreviations: AC, anthracycline and cyclophosphamide; 5-HT₃, 5-hydroxytryptamine-3; NK₁, neurokinin 1.

Penggunaan 5-HT3 pada RINV

MASCC/ESMO 2024:

High risk (mis. TBI): 5-HT3 antagonist + dexamethasone

Moderate risk (mis. upper abdomen/craniospinal): 5-HT3 antagonist ± dexamethasone

Low & minimal risk: tidak rutin profilaksis, gunakan rescue (dexamethasone /
dopamine-antagonist / 5-HT3 antagonist)

ASCO guideline (RINV) 2011:

High risk RT: 5-HT3 antagonist sebelum tiap sesi & lanjut ≥24 jam setelah selesai RT

Table 3. Antiemetic Dosing by Chemotherapy Risk Category

Risk Category	Dosing on Day of Chemotherapy	Dosing on Subsequent Days
High emetic risk*		
NK ₁ antagonist		
Aprepitant	125 mg oral	80 mg oral; days 2 and 3
Fosaprepitant	150 mg IV	
5-HT ₃ antagonist		
Granisetron	2 mg oral; 1 mg or 0.01 mg/kg IV	
Ondansetron	8 mg oral twice daily; 8 mg or 0.15 mg/kg IV	
Palonosetron	0.50 mg oral; 0.25 mg IV	
Dolasetron	100 mg oral ONLY	
Tropisetron	5 mg oral; 5 mg IV	
Ramosetron	0.3 mg IV	
Corticosteroid if aprepitant is used†		
Dexamethasone	12 mg oral or IV	8 mg oral or IV; days 2 and 3 or days 2-4
Corticosteroid if fosaprepitant is used†		
Dexamethasone	12 mg oral or IV	8 mg oral or IV day 2; 8 mg oral or IV twice per day on days 3 and 4
Moderate emetic risk‡		
5-HT ₃ antagonist		
Palonosetron	0.50 mg oral; 0.25 mg IV	
Corticosteroid		
Dexamethasone	8 mg oral or IV	8 mg; days 2 and 3
Low emetic risk		
Corticosteroid		
Dexamethasone	8 mg oral or IV	

https://www.oegari.at/web_files/cms_dat/en/esa_asa_2020_consensus_guidelines_pov.pdf

Dosis Tropisetron pada RINV

tropisetron 5 mg oral atau IV

Apa itu Tropisetron

Obat anti-mual/anti-muntah/antiemetik dari golongan antagonis reseptor serotonin 5-HT₃

Antagonis = Penjahat yang menahan sinyal

Reseptor = Penangkap sinyal

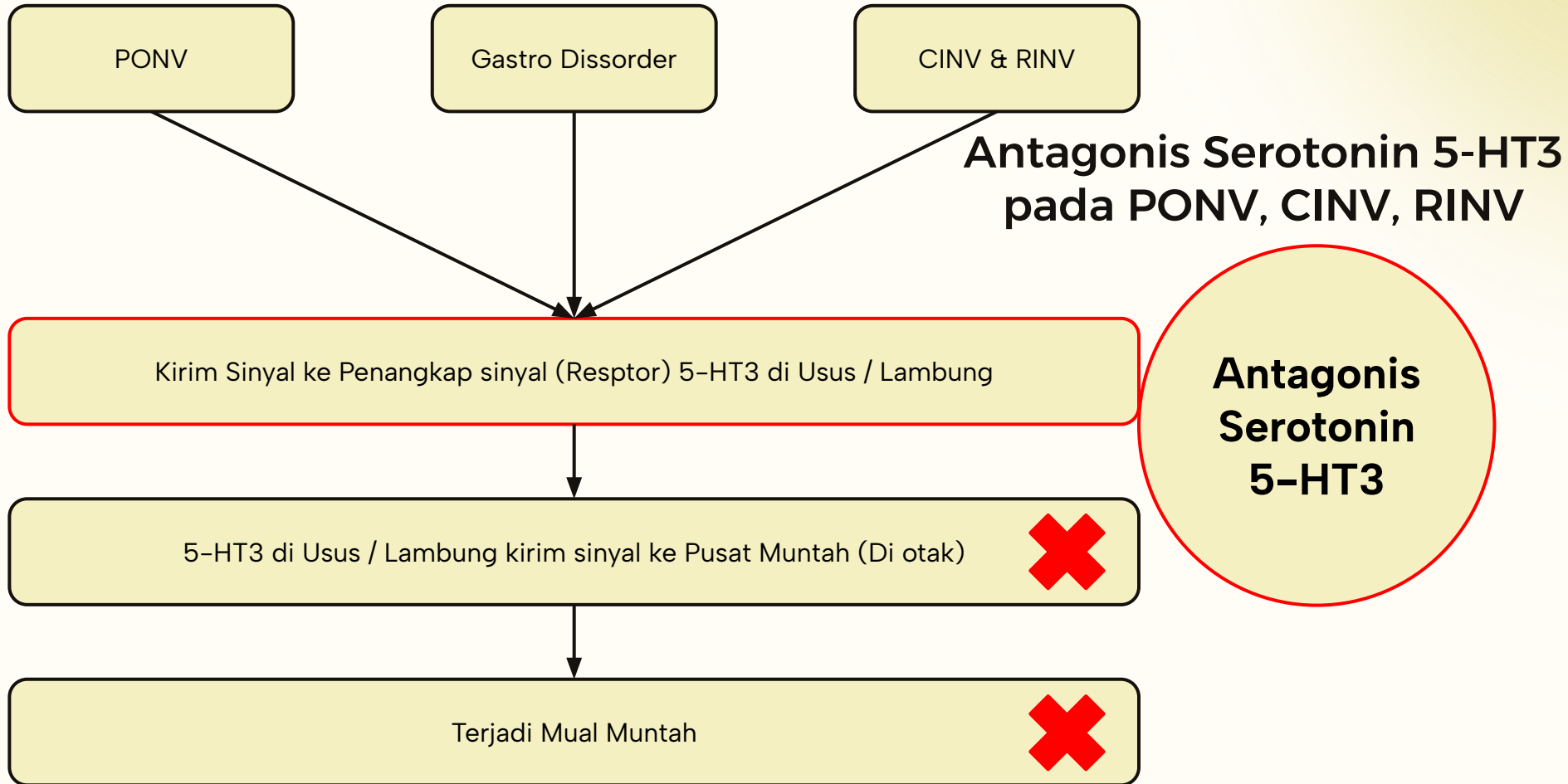
5-HT₃ = Pengirim sinyal muntah ke pusat muntah di otak

Cara kerjanya

Tropisetron sebagai antagonis → nempel penangkap sinyal (reseptor) 5-HT₃ (serotonin)

Serotonin nggak bisa kirim sinyal ke pusat muntah!

Mual / Muntah ↓↓↓



Apa Saja Obat Golongan 5-HT3 Anatagonis?

Ondansetron

Granisetron

Tropisetron

Dolasetron

Palonosetron

Ramosetron

Awal 1990-an:
Kelompok 5-HT3 antagonist mulai
muncul
Ondansetron, granisetron,
tropisetron

Setelah itu dolasetron

Pemakaian makin luas untuk
CINV/PONV/RINV

2000-an:
palonosetron sebagai generasi lebih
baru

Dosis Tropisetron

Indikasi	Maksimal yang relevan secara klinis
PONV, CINV & RINV	2 – 5 mg/hari, 1x sehari
Anak >2 tahun	0,2 mg/kg/hari, maksimal 5 mg/hari
Pasien hipertensi tidak terkontrol	hindari >10 mg/hari → menaikkan tekanan darah
Dosis tertinggi (data klinis label)	40 mg IV/hari (bukan dosis terapi rutin)

Tropisetron Pemakaian di IGD

Banyak pasien datang ke IGD atau dirawat bukan karena kanker/operasi, tapi karena mual-muntah akibat gangguan saluran cerna

Gastroenteritis akut
Gastritis/dispepsia berat
Distensi lambung/ileus ringan
Efek obat, termasuk opioid
Pasien rawat inap yang sulit makan/minum karena mual



5-HT3 serotonin
dari saluran cerna
mengirim sinyal
ke pusat muntah



Tropisetron
hambat
pengiriman sinyal
Tersebut

Penyebab Gastro Memicu Jalur 5-HT₃

Iritasi mukosa usus/lambung
Infeksi, toksin makanan, inflamasi, obat ngiritasi saluran cerna.

Peregangan lambung/usus
Perut kembung, distensi, ileus ringan, atau perlambatan pengosongan lambung dapat memicu sinyal mual

Serotonin dari sel usus
Lapisan usus teriritasi → melepas serotonin.

Perbedaan Obat Colongan 5-HT3 Anatagonis

Obat	Lama bertahan (t½)	Keterangan
Ondansetron	3–6 jam	Durasi relatif pendek → kadang butuh strategi dosis/interval
Granisetron	8–9 jam	Umumnya lebih lama dari ondansetron
Tropisetron	+–8 jam	Durasi menengah
Palonosetron	40 jam	Cocok untuk cakupan lebih panjang (delayed mual muntah)
Ramosetron	9 jam	Durasi relatif panjang

Tropisetron VS Metoclopramide

Metode: RCT double-blind
Populasi: 100 pasien dewasa di IGD/ED dengan mual-muntah

Intervensi:
Tropisetron 5 mg IV bolus vs metoclopramide 10 mg IV bolus

Hasil:
Muntah hingga menit ke-180: 2/50 pasien, 4,0% pada tropisetron vs 9/50 pasien, 18,0% pada metoclopramide, $p = 0,05$
Kebutuhan rescue antiemetic: 5/50, 10,0% pada tropisetron vs 13/50, 26,0% pada metoclopramide
Akathisia lebih rendah pada kelompok tropisetron dibanding metoclopramide

Tropisetron lebih baik daripada metoclopramide dalam menurunkan kejadian muntah objektif pada pasien IGD dengan mual-muntah umum

Table 2. Differences in mean nausea scores from baseline versus time, mean (SD)

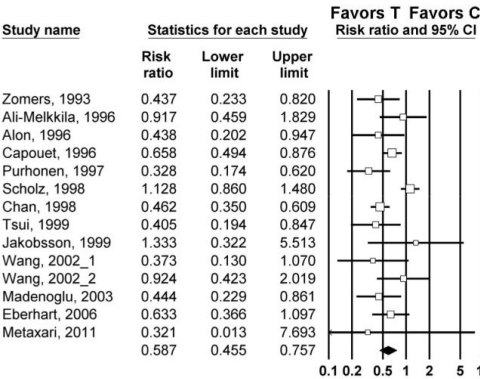
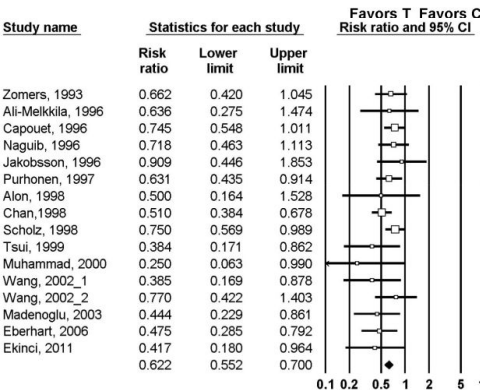
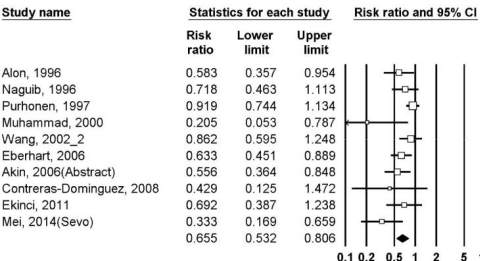
Time point		Tropisetron ($n = 50$)†	Metoclopramide ($n = 50$)†	Difference in means (95% CI)
0	Mean (SD)	baseline	baseline	baseline
30	Mean (SD)	-25.2 (25.4)	-26.4 (19.7)	1.2 (-8.5-10.9)
60	Mean (SD)	-37.7 (22.8)	-29.4 (24.8)	8.3 (-2.1-18.7)
90	Mean (SD)	-42.5 (24.9)	-34.7 (25.3)	7.8 (-3.8-19.3)
120	Mean (SD)	-44.4 (24.0)	-33.7 (24.6)	10.7 (-0.3-21.7)
180	Mean (SD)	-47.9 (22.3)	-37.0 (25.4)	10.9 (-0.7-22.6)

†Sample size varies between groups and times, from 32 to 50. This is due to patients being unavailable within the ED (e.g. in radiology) or being discharged within the data collection period.

Table 3. Mean akathisia scores versus time, mean (SD)

Time point		Tropisetron ($n = 50$)†	Metoclopramide ($n = 50$)†	Difference in means (95% CI)
0	Mean (SD)	0.06 (0.31)	0.38 (1.84)	0.3 (-0.2-0.8)
30	Mean (SD)	0.57 (1.44)	1.70 (3.07)	1.1 (0.1-2.2)
60	Mean (SD)	0.39 (1.07)	1.62 (3.78)	1.2 (0.01-2.5)
90	Mean (SD)	0.19 (0.66)	0.79 (2.26)	0.6 (-0.2-1.4)
120	Mean (SD)	0.09 (0.38)	0.81 (2.12)	0.7 (-0.02-1.5)
180	Mean (SD)	0.07 (0.38)	0.17 (0.91)	0.1 (-0.3-0.5)

†Sample size varies between groups and times, from 32 to 50. This is due to patients being unavailable within the ED (e.g. in radiology) or being discharged within the data collection period.



Tropisetron pada PONV

Metode: Systematic review dan meta-analysis dari RCT

Populasi & Intervensi:

30 RCT dengan total 5175 pasien dewasa yang menjalani operasi dengan general anesthesia
Tropisetron profilaksis perioperatif

Hasil:

Tropisetron menurunkan kejadian mual pascaoperasi secara signifikan.
RR 0,718

Tropisetron menurunkan kejadian muntah pascaoperasi secara signifikan.
RR 0,587

Tropisetron menurunkan kejadian gabungan mual dan muntah pascaoperasi secara signifikan.
RR 0,655

Tropisetron menurunkan kebutuhan obat antiemetik tambahan secara signifikan.
RR 0,622

Tropisetron lebih baik daripada metoclopramide dalam menurunkan kejadian muntah objektif pada pasien IGD dengan mual-muntah umum

Tropisetron + Dexamethasone pada Anestesi Remimazolam

Metode: RCT double-blind
Populasi: 192 pasien operasi day surgery ginekologi (anestesi remimazolam + alfentanil)

Intervensi: Semua dapat dexamethasone 5 mg i.v. lalu:
Grup DT: tambah tropisetron 5 mg i.v.
Kontrol DC: tambah salin

Hasil:
PONV di PACU: DT 26,7% vs DC 50%, $p < 0,01$
PONV 24 jam: DT 45,1% vs DC 63,8%, $p > 0,05$

Tropisetron + deksametason lebih baik untuk PONV fase awal (Post Anesthesia Care Unit/PACU)

Table 2 Postoperative values

PONV	DD group	DT group	DC group	F/ χ^2 /H(K)	P value
PACU(n)	62	60	58		
0	53(85.5%)	44(73.3%)	29(50.0%)	18.443	0.000
1	9(14.5%)	16(26.7%)	29(50.0%) ^a		
24 h(n)	47	51	47		
0	26(55.3%)	28(54.9%)	17(36.2%)	4.558	0.102
1	21(44.7%)	23(45.1%)	30(63.8%)		

PONV Postoperative nausea and vomiting PACU Postanesthesia care unit

0 = no of nausea 1 = nausea or/and vomiting

^a $P < 0.01$ for DC group vs DD group and DT group

Tropisetron + Dexamethasone pada Laparoskopi Ginekologi

Metode: RCT

Populasi: 157 pasien laparoskopi ginekologi

Intervensi:

Trp: tropisetron + dexamethasone (kombinasi)

Dxm: dexamethasone saja

Hasil 24 jam:

PONV: Trp 26% vs Dxm 50%

Beda signifikan: $p = 0,042$ (OR 0,359; 95%CI 0,157–0,819)

Menambah tropisetron ke dexamethasone menurunkan PONV 24 jam pada laparoskopi ginekologi

Table 2 Postoperative nausea and vomiting in the first day (0–24 h), patients' needs for rescue antiemetics, and patient satisfaction scores. Values are median (minimum–maximum) or n (%; 95% CI). Group Acu, P6 acustimulation and dexamethasone group; Group Dxm, dexamethasone alone group; Group Trp, tropisetron and dexamethasone group; PONV, postoperative nausea and vomiting. *P-value represents χ^2 analysis and †P-value refers to Kruskal–Wallis H-test. ‡Significant difference vs Group Dxm with Holm–Bonferroni correction

Variable	Group Acu (n=50)	Group Trp (n=53)	Group Dxm (n=50)	P-value
0–24 h PONV				
Nausea, vomiting, or both [incidence; n (%; 95% CI)]	14 (28, 15–41) [‡]	14 (26, 14–39) [‡]	25 (50, 36–64)	0.021*
Nausea [incidence; n (%; 95% CI)]	14 (28, 15–41) [‡]	14 (26, 14–39) [‡]	25 (50, 36–64)	0.021*
Vomiting [incidence; n (%; 95% CI)]	9 (18, 7–29)	8 (15, 5–25)	12 (24, 12–36)	0.503*
Need for rescue antiemetics				
Metoclopramide [proportion; n (%; 95% CI)]	5 (10, 1–19)	4 (8, 0–15)	7 (14, 4–24)	0.560*
Metoclopramide [mg; median (range)]	0 (0–50)	0 (0–20)	0 (0–40)	0.581 [†]
Patient satisfaction [median (range)]	95 (85–100)	92 (80–100)	94 (85–100)	0.230 [†]

Tropisetron vs Dexamethasone Vs Kombinasi pada Thyroidectomy

Metode: RCT
Populasi: 150 pasien thyroidectomy (50/grup)

Intervensi:
D: dexamethasone 8 mg
T: tropisetron 5 mg
D+T: kombinasi 8 mg + 5 mg

Hasil:
Complete response (tidak mual & tidak muntah): D+T 78% (39/50) vs D 58% (29/50) vs T 66% (33/50); $p = 0,01$

Kombinasi tropisetron+dexamethasone lebih kuat untuk cegah PONV (terutama 6–48 jam)

Table 2 Postoperative events

Event	D (n = 50)	T (n = 50)	D+T (n = 50)	P
Complete response	29 (58%)	33 (66%)	39 (78%)	0.01
Nausea	21 (42%)	17 (34%)	11 (22%)	0.01
Vomiting	16 (32%)	15 (30%)	8 (16%)	<0.05
Nausea events	64	56	33	<0.01
Early (0–6 h)	19	23	12	NS
Late (6–48 h)	45	33	21	<0.01
Vomiting events	37	40	16	<0.01
Early (0–6 h)	11	14	5	NS
Late (6–4 h)	26	26	11	<0.05
Antiemetic consumption				
Metoclopramide use				
No. of patients	16	15	10	NS
Total dose (mg)	160	150	100	NS
Tropisetron use				
No. of patients	7	8	3	NS
Total dose (mg)	35	40	20	NS
Analgesic consumption	0	0	0	NS

Tropisetron Dosis Kecil pada Anak Operasi Strabismus

Metode: RCT placebo-controlled
Populasi: 125 anak usia 2–12 tahun, operasi strabismus

Intervensi: Setelah induksi anestesi: placebo vs tropisetron 0,5 / 1 / 1,5 / 2 mg/m² (single dose)

Hasil:
Muntah pasca operasi (2, 6, 24 jam): placebo lebih tinggi, beda signifikan $p < 0,001$
Antar dosis tropisetron: tidak beda bermakna ($p > 0,05$); angka POV di kelompok tropisetron sekitar 16–24%

Pada anak, tropisetron single dose menurunkan muntah awal, dan dosis kecil sudah cukup

Randomized Controlled Trial > Mt Sinai J Med. 2006 Dec;73(8):1106-11.

A randomized, placebo-controlled trial of a single dose of tropisetron for the prevention of vomiting after strabismus surgery in children

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Abstract

Purpose: This study evaluates the effect of different doses of tropisetron to prevent postoperative vomiting, which frequently occurs in children following strabismus surgery.

Methods: One hundred twenty-five American Society of Anesthesiologists' classification (ASA) I-II group patients 2-12 years of age were randomized to this prospective, single-blind, placebo-controlled clinical study. Patients were placed in groups of 5 and did not receive any premedication. The first group received placebo, and the 2nd, 3rd, 4th, and 5th groups received 0.5 mg/m(2), 1 mg/m(2), 1.5 mg/m(2), and 2 mg/m(2) of tropisetron, respectively, following anesthesia induction. The same anesthetic technique and analgesia were used for all groups. The patients were examined for the presence of vomiting and for any complaints and side effects at 2, 6, and 24 hours after surgery.

Results: The incidence of postoperative vomiting (POV) was statistically more significant in the placebo group at 2, 6 and 24 hours, when compared to the study groups ($p < 0.001$), but there was no significant difference among tropisetron groups at 6-24 hours ($p > 0.05$). There was no significant difference in terms of the incidence of POV among the study groups (16%, 16%, 24%, 20% respectively) at all periods ($p > 0.05$). The number of patients with POV score of 3 was 10 in the placebo group, while it was 1, 2, 0 and 1 in the 2nd, 3rd, 4th, and 5th groups, respectively ($p < 0.01$).

Conclusions: Tropisetron (0.5, 1.0, 1.5 and 2.0 mg/m(2)) decreased the incidence and severity of POV following strabismus surgery in children. All of the doses seemed to be equally effective. There was no difference in POV control between placebo and any of the doses of the tropisetron after six hours. So we suggest that 0.5 mg/m(2) single-dose tropisetron is enough for preventing POV following strabismus surgery in children.

Tropisetron pada Pasien Craniotomy

Metode: RCT double-blind placebo-controlled
Populasi: 65 pasien craniotomy supratentorial

Intervensi: tropisetron 2 mg i.v. saat penutupan dura vs placebo

Hasil (24 jam):
Nausea: 30% vs 46,7% ($p > 0,05$)
Emetic episodes (muntah/episode muntah): 26,7% vs 56,7% ($p < 0,05$)
Butuh rescue antiemetik: 26,7% vs 60% ($p < 0,05$)

Tropisetron 2 mg jelas menurunkan muntah dan kebutuhan rescue pasca craniotomy

TABLE 2. Incidence of nausea and vomiting and rescue antiemetic medication

	Nausea		Emetic episode	
	Group P (n = 30)	Group T (n = 30)	Group P (n = 30)	Group T (n = 30)
0-2 h	14 (46.7)	9 (30)	14 (46.7)	6 (20)
2-12 h	6 (20)	5 (16.7)	6 (20)	6 (20)
12-24 h	4 (13.3)	0	1 (3.3)	2 (6.67)
Overall 24 h	14 (46.7)	9 (30)	18 (60)	8 (26.7)*
Antiemetics			18 (60)	8 (26.7)*

Data are no. (%) of patients.
P = placebo; T = tropisetron.
* $P < .05$.

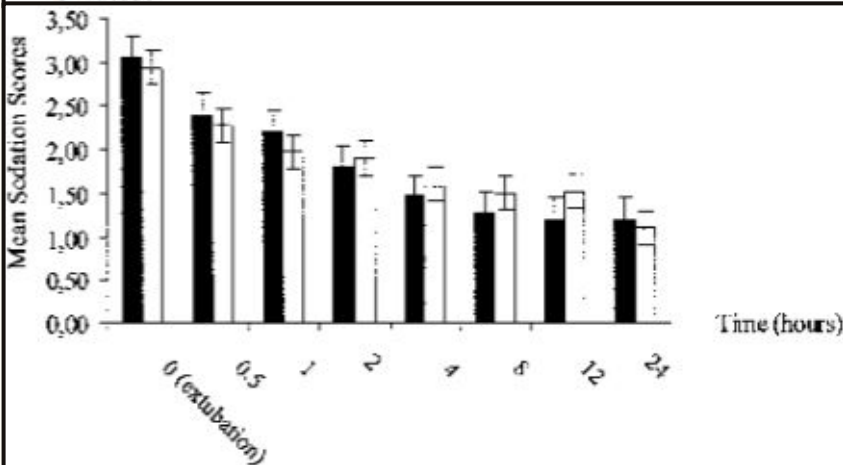


FIG. 1. Mean sedation scores of two groups.

Tropisetron 2 mg vs 5 mg (PONV)

Metode: RCT double-blind
placebo-controlled
Populasi: 60 perempuan (3 grup
@20), operasi anestesi umum

Intervensi: Saat induksi: placebo vs
tropisetron 2 mg vs 5 mg i.v. (single
dose)

Hasil:
Muntah 0–2 jam: placebo 12/20
(60%) vs trop 2 mg 1/20 (5%) vs
trop 5 mg 1/20 (5%) ($P < 0,05$)

**Tropisetron 2 mg tampak setara
efektif dengan 5 mg untuk cegah
muntah dini pasca operasi**

TABLE 2:

Incidence of vomiting and rescue treatment in surgical patients randomized to receive either saline (control) or 2 mg or 5 mg intravenous tropisetron for the prevention of post-operative nausea and vomiting

	Group 1 (saline)	Group 2 (2 mg tropisetron)	Group 3 (5 mg tropisetron)	<i>P</i> -value
Incidence at 2 h post-operation:				
Vomiting, <i>n</i> (%)	12/20 (60)	1/20 (5)	1/20 (5)	$P < 0.002$ for group 1 versus group 2 and group 1 versus group 3
Rescue therapy, <i>n</i> (%)	0/20	3/20 (15)	3/20 (15)	–
Incidence at 2 – 4 h post-operation:				
Vomiting, <i>n</i> (%)	4/20 (20)	0/20	0/20	$P < 0.05$ for group 1 versus group 2 and group 1 versus group 3
Rescue therapy, <i>n</i>	0/20	0/20	0/20	–

Tropisetron Profilaksis pada Operasi Ginekologi

Metode: RCT placebo-controlled
Populasi: 80 perempuan operasi ginekologi (40 tropisetron vs 40 placebo)

Intervensi: Tropisetron (single dose i.v.) vs placebo

Hasil (24 jam):
Muntah: 17,5% (7/40) vs 40% (16/40); $p < 0,05$
Mual: 30% (12/40) vs 52% (21/40); $p < 0,05$
Total effective response: 65% (26/40) vs 40% (16/40); $p < 0,05$

Tropisetron menurunkan mual & muntah pasca operasi ginekologi

Table 2. Postoperative Rate of Nausea and Emesis During the 24 h After Recovery in the Tropisetron and Placebo Groups

	Tropisetron (n = 40)	Placebo (n = 40)	Statistical significance
No nausea and no emesis	26 (65%)	16 (40%)	$P = 0.02$
Nausea and/or emesis	14 (35%)	24 (60%)	$P < 0.02$
Nausea all grades (mild, moderate, severe)	12 (30%)	21 (52%)	$P < 0.04$
Nausea first 4 h	10 (25%)	19 (47%)	$P < 0.03$
Nausea last 20 h	3 (7%)	4 (10%)	NS
Emesis (vomiting and/or retching)	7 (17%)	16 (40%)	$P < 0.02$
Emesis first 4 h	6 (15%)	12 (30%)	NS
Emesis last 20 h	1 (2%)	6 (15%)	$P < 0.05$

Values refer to number of patients (percentage in brackets).

NS = not significant.

Tropisetron + Dexamethasone vs Tropisetron Saja (Cisplatin)

Metode: RCT double-blind multicenter parallel
Populasi: 282 pasien kanker, chemo-naïve, high-dose cisplatin

Intervensi:
Tropisetron 5 mg i.v. hari-1 lalu 5 mg oral hari 2–6
Ditambah dexamethasone vs placebo

Hasil:
Complete protection akut (hari-1): muntah 76,3% vs 55,2%; mual 79,3% vs 61,5%
Complete protection delayed (hari 2–6): muntah 60% vs 39,2%; mual 60% vs 40,6%

Menambah dexamethasone ke tropisetron meningkatkan kontrol emesis akut & delayed pada cisplatin

Table 2. Anti-emetic efficacy on acute and delayed nausea and vomiting

	Tropisetron plus placebo (n = 143) % of patients (95% confidence interval)	Tropisetron plus dexamethasone (n = 135) % of patients (95% confidence interval)	
Acute emesis			
Complete protection from vomiting	55.2 (47.1–63.4)	76.3 (69.1–83.5)	0.0002
Complete protection from nausea	61.5 (53.6–69.5)	79.3 (72.4–86.1)	0.001
Complete protection from both vomiting and nausea	46.2 (38.0–54.3)	71.1 (63.5–78.8)	0.0002
Treatment failure	10.5 (5.5–15.5)	7.4 (3.6–13.2)	0.37
Delayed emesis (days 2–6)			
Complete protection from vomiting	39.2 (31.2–47.2)	60.0 (51.7–68.3)	0.0005
Complete protection from nausea	40.6 (32.5–48.6)	60.0 (51.7–68.3)	0.001
Complete protection from both vomiting and nausea	26.6 (19.3–33.8)	51.1 (42.7–59.5)	0.0001
Treatment failure	3.9	3.0	0.32

Tropisetron vs Standard Therapy pada Pasien CINV Refrakter

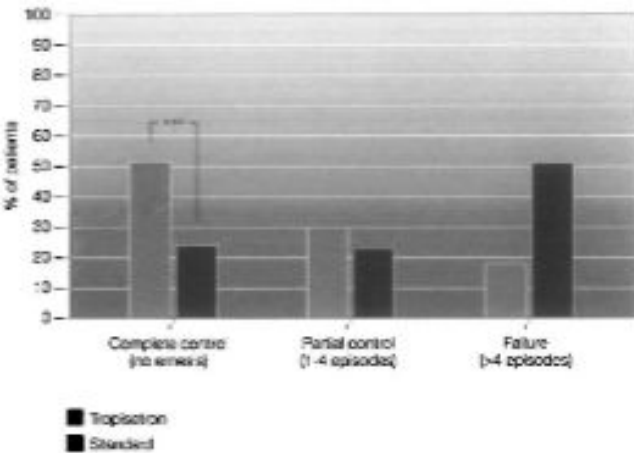


Fig 1. Control of emesis on the first day of chemotherapy (%).
*** $p < 0.001$.

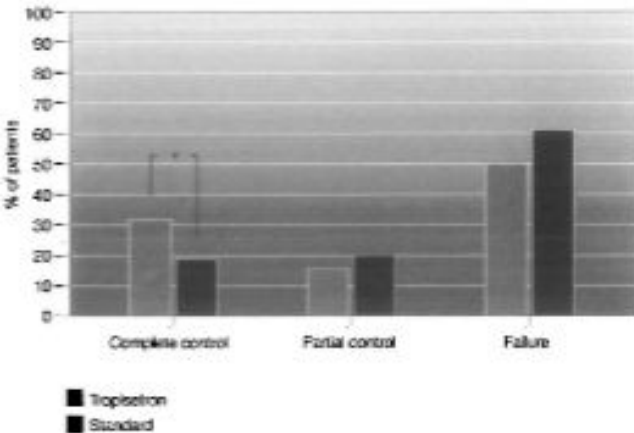


Fig 2. Control of nausea on the first day of chemotherapy (%).
* $p < 0.05$.

Metode: RCT multicenter

Populasi: 231 pasien kanker yang gagal antiemetik konvensional sebelumnya (trop $n=115$ vs standar $n=116$)

Intervensi: Tropisetron 5 mg/hari vs terapi standar optimal (tanpa 5-HT3 antagonist)

Hasil (hari-1):

Acute vomiting terkontrol: 52% (60/115) vs 25% (29/116); $p < 0,001$

Acute nausea hilang total: 32% (37/115) vs 19% (22/116); $p < 0,05$

Extrapyramidal symptoms: 0% (tropisetron) vs 13% (standar)

Buat pasien yang ga ampu sama obat lama, pindah ke tropisetron lebih efektif daripada mengoptimalkan obat standar

Table 5. Efficacy analysis of course 1. Proportion of patients showing control of vomiting, nausea, and vomiting and nausea combined. Results are presented for each of days 1–6 and for the entire course, based on a worst-day analysis

Day	1	2	3	4	5	6	Overall
Vomiting							
Total control (%)	53	43	58	79	81	83	28
Partial control (%)	31	30	33	13	14	13	34
No control (%)	15	26	6	4	1	0	34
Not assessed (%)	1	1	3	4	4	4	3
Nausea							
Total control (%)	47	27	37	43	54	62	15
Partial control (%)	48	54	46	45	38	34	59
No control (%)	4	18	15	8	4	0	23
Not assessed (%)	1	1	3	4	4	4	3
Vomiting and nausea combined							
Complete response (%)	38	23	32	41	53	61	12
Partial response (%)	43	44	48	44	38	36	39
No response (%)	19	31	17	11	6	0	39
Not assessed (%)	1	1	3	4	4	4	10

Not assessed refers to patients who prematurely discontinued the treatment.

Tropisetron + Dexamethasone vs Tropisetron Saja (Cisplatin)

Metode: RCT multicenter double-blind

Intervensi: Tropisetron + dexamethasone vs tropisetron + placebo

Hasil:

Total control acute nausea: 75% vs 37%, $p = 0,001$

Acute vomiting prevented: 75% vs 40%, $p = 0,001$

Dexamethasone → booster yang kuat untuk membuat tropisetron lebih sukses pada CINV cisplatin

Tropisetron 2 mg vs 5 mg untuk CINV

Metode: RCT double-blind multicenter

Populasi: 152 pasien chemo-naïve (dievaluasi cycle-1; ITT 144, evaluable 124)

Intervensi: Tropisetron 2 mg vs 5 mg (i.v. day-1, oral day 2-6)

Hasil:

Total control acute vomiting: 73% (5 mg) vs 55% (2 mg); $p = 0,02$

Total control acute nausea: 70% (5 mg) vs 51% (2 mg); $p = 0,03$

Headache lebih sering di 5 mg: $p < 0,05$

CINV (low-dose cisplatin/non-cisplatin), 5 mg lebih kuat di fase akut, tapi lebih sering sakit kepala

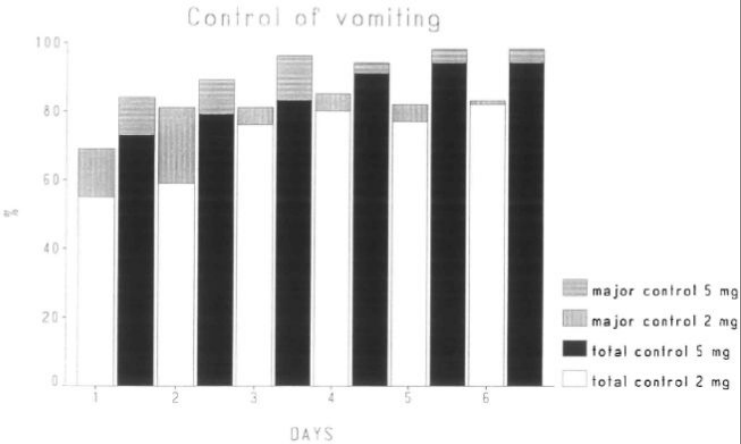


Figure 1. Proportion of ITT-patients with total and major control of vomiting on each of days 1-6.

Table 3. Incidences of adverse events.

	Tropisetron dose		P-value
	2 mg (n = 74)	5 mg (n = 70)	
No of patients reporting adverse events	43 (58%)	49 (70%)	NS
Headache	12 (16%)	25 (36%)	<0.05
Constipation	8 (11%)	13 (19%)	NS
Abdominal pain	5 (7%)	7 (10%)	NS
Fatigue	3 (4%)	7 (10%)	NS
Dizziness	7 (9%)	2 (3%)	NS
Rigors	5 (7%)	4 (6%)	NS
Anorexia	4 (5%)	4 (6%)	NS
Insomnia	4 (5%)	2 (3%)	NS
Malaise	4 (5%)	2 (3%)	NS
Pain	1 (1%)	4 (6%)	NS

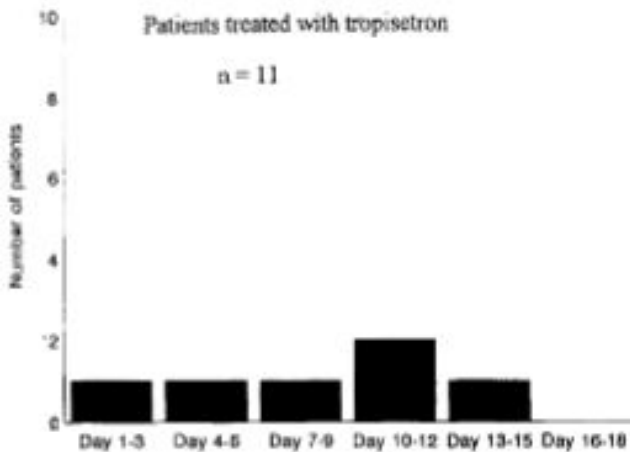
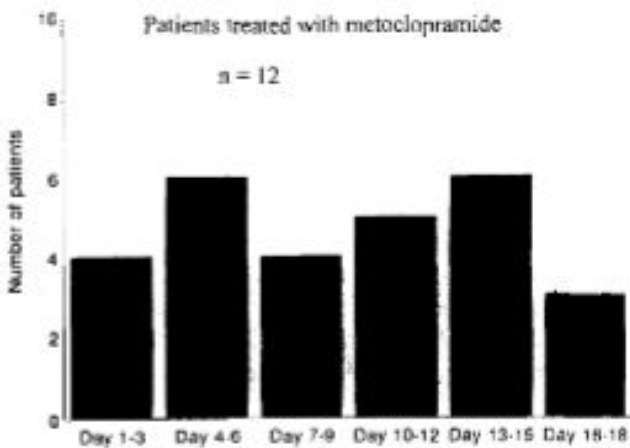


Fig. 1. Number of patients with 'significant nausea' (score >25 mm on the visual analogue scale) within each 3-day period.

Tropisetron Profilaksis saat Radioterapi Abdomen Seminoma

Metode: RCT open-label
Populasi: 88 pasien radioterapi (abdomen/lapangan besar)

Intervensi:
Tropisetron 5 mg oral/hari (n=11) vs metoclopramide 30 mg oral/hari (n=12; bisa dinaikkan)

Table 1

Gastrointestinal toxicity in patients with seminoma stage I during abdominal radiotherapy

	MET group ^a (n = 12)	TROP group ^b (n = 11)
Nausea*	1.32 ^c (0-4.60) ^d	0.14 (0-2.55)
Significant nausea* (at least once)	9 ^e	2
>0 emetic event (mean)**	6	1
Abdominal pain	0.14 0-2.00	0.03 0-0.78
Bowel motion	1.58 0.72-4.72	2.06 0.89-6.78
Premature discontinuation	1	3 ^f

*Patients treated with metoclopramide.

^bPatients treated with tropisetron.

^cMedian.

^dRange: increasing figures indicate increasing symptomatology.

^eNumber of patients with mean >0.

^fFailure: n = 2; side effects: n = 1

*P < 0.05; **P = 0.07.

Hasil:
Awal: metoclopramide 10 mg oral 3x/hari bila muncul mual berat
Jika gagal: rescue tropisetron 5 mg oral 1x/hari

Tropisetron lebih baik menekan mual pada radioterapi abdomen (dan tren lebih baik untuk muntah)

TRopisetron vs 5-HT3 Antagonist

5-HT3	Biasa dipakai	Kapan Pakai TRopisetron
Ondansetron	Paling familiar & tersedia luas	PONV profilaksis / CINV akut
Granisetron	RS sudah pakem sama granisetron	PONV & CINV akut
Palonosetron	Fokus coverage panjang (yang delayed)	Fokus fase akut / kebutuhan RS langsung (IV)
Dolasetron	(Jarang dipakai)	PONV / CINV akut
Ramosetron	(Jarang dipakai)	PONV / CINV akut

PONV: Laparoscopic Cholecystectomy (Tropisetron vs Granisetron)

Metode: RCT, double-blind, dibanding placebo
Populasi: 132 pasien ASA I-II operasi lap chole (dinilai 0–24 jam)

Intervensi: Tropisetron 5 mg IV (infus pendek, ±10 menit sebelum induksi)

Hasil:

Kejadian PONV: 52% (13/25) tropisetron vs 48% (12/25) granisetron vs 72,4% (21/29) placebo
Statistik: tidak ada perbedaan bermakna antara ondansetron–tropisetron–granisetron; ondansetron lebih rendah vs metoklopramid/placebo ($P = 0,02$)
Emesis-free 24 jam: 48% tropisetron vs 52% granisetron vs 27,6% placebo

Tropisetron dan granisetron hasilnya mirip

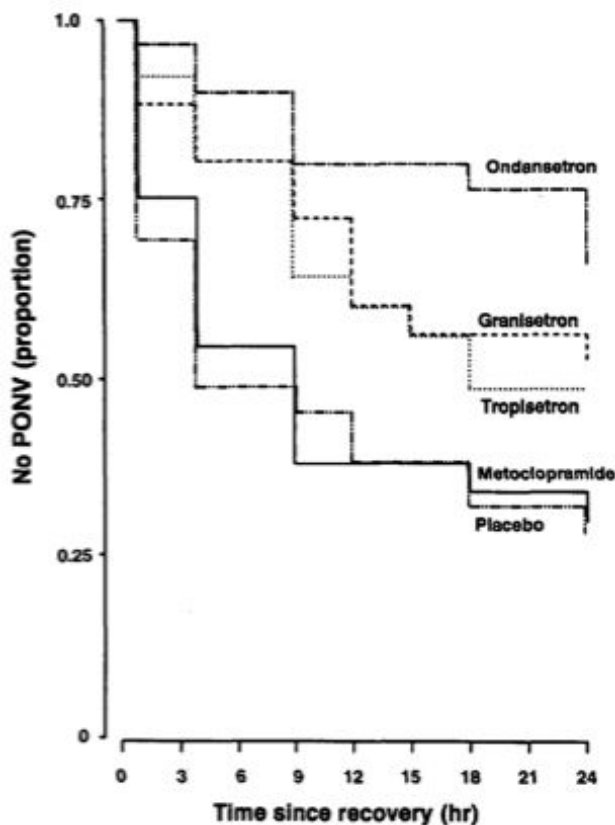


FIGURE “Survival” curves for the ondansetron 4 mg, tropisetron 5 mg, granisetron 3 mg, metoclopramide 10 mg or placebo (normal saline) groups. Proportion of patients in each group who had no PONV and had not required any rescue antiemetic therapy after recovery from anaesthesia. The times at which rescue antiemetic was first received were longer ($P < 0.01$) in ondansetron group than in the placebo and metoclopramide groups. There was no difference among the ondansetron, granisetron and tropisetron groups.

TABLE II Incidence of postoperative nausea and vomiting (PONV)

	Ondansetron (4 mg)	Tropisetron (5 mg)	Granisetron (3 mg)	Metoclopramide (10 mg)	Placebo (NaCl 0.9%)
n	29	25	25	24	29
Patients with PONV (%)	10 (34.5%)	13 (52%)	12 (48%)	17 (70.8%)	21 (72.4%)

The incidence of no PONV was lower in the ondansetron group as compared with metoclopramide and placebo groups ($P = 0.02$). There were no differences among ondansetron, tropisetron and granisetron groups.

Comparison of granisetron, ondansetron, and tropisetron in the prophylaxis of acute nausea and vomiting induced by cisplatin for the treatment of head and neck cancer: a randomized controlled trial

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PMID: 8608488

Abstract

Background: A single-institution, prospective, randomized, open controlled trial was carried out on head and neck cancer patients to compare granisetron (GRA), ondansetron (OND), and tropisetron (TRO) in the prevention of cisplatin-induced acute nausea and vomiting. All patients were chemotherapy-naïve and treated with cisplatin on Day 1 (80 to 100 mg/m²).

Methods: One hundred seventeen patients were treated for a total of 463 cycles of cisplatin-based chemotherapy and randomized to receive 24 mg of OND intravenously (i.v.), 3 mg of GRA i.v., or 5 mg of TRO i.v. for the control of acute nausea and emesis.

Results: In the GRA group, complete response (CR) was obtained in 119 of 165 cycles (72.1%), major response (MR) in 32 cycles (19.4%), minor response (MiR) in 5 cycles (3%), and a failure (F) in 9 cycles (5.5%). In the OND group, CR was obtained in 110 of 150 cycles (73.3%), MR in 31 cycles (20.7%), MiR in 2 cycles (1.3%), and F in 7 cycles (4.7%). In the TRO group, CR was obtained in 100 of 148 cycles (67.6%), MR in 26 cycles (17.6%), MiR in 15 cycles (10.1%), and F in 7 cycles (4.7%). Major efficacy (CR + MR) was obtained in 151 of 165 cycles (91.5%) for GRA, in 141 of 150 cycles (94.0%) for OND, and in 126 of 148 cycles (85.2%) for TRO. The difference in major efficacy between OND and TRO was statistically significant. When comparing MiR, both GRA and OND were more effective than TRO. No other significant differences were observed among the three antiemetic agents.

Conclusions: Although our results were achieved in an open trial, they show that GRA and OND are equally effective antiemetic agents in the prevention of cisplatin induced acute nausea and vomiting. TRO provides almost the same protection but is not as effective as OND for major efficacy. All three antiemetics can be administered safely to patients undergoing chemotherapy with cisplatin at doses of 80 mg/m² or more.

CINV (Acute): Cisplatin Head & Neck Cancer (Tropisetron vs Granisetron)

Metode: RCT (open controlled), analisis per siklus kemo
Populasi: 117 pasien kanker kepala-leher; total 463 siklus cisplatin (80–100 mg/m²)

Intervensi: Tropisetron 5 mg IV pada hari cisplatin (profilaksis fase akut)

Hasil (fase akut):
Complete response (CR): 67,6% (100/148 siklus) tropisetron vs 72,1% (119/165) granisetron (P = 0,451)
Major efficacy (CR+MR): 85,2% (126/148) tropisetron vs 91,5% (151/165) granisetron (P = 0,112)
ondansetron vs tropisetron untuk CR+MR (P = 0,021)

Tropisetron dan granisetron hasilnya mirip juga!

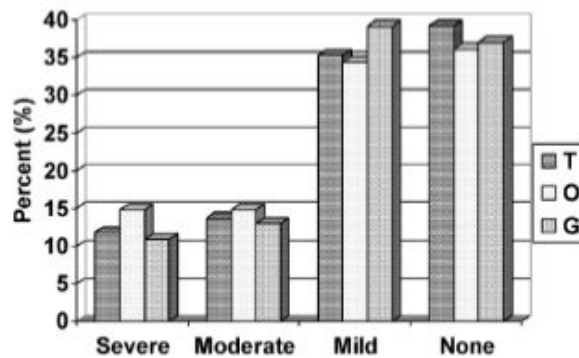


Figure 4. Observed frequencies of nausea of highest rank according to tropisetron (T), ondansetron (O), and granisetron (G) within first 24 hours following chemotherapy administration. There were no differences in response rates among the 3 5HT₃ receptor antagonists ($p = 0.995$).

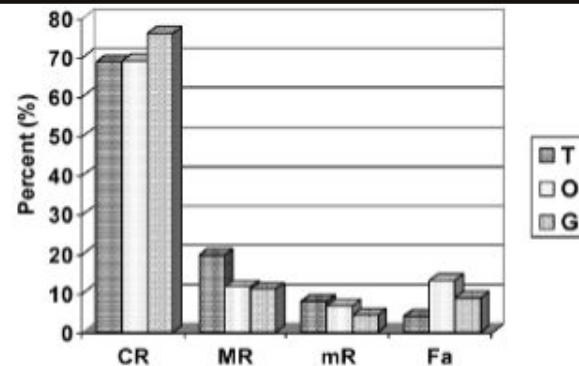


Figure 3. Observed frequencies of responses to tropisetron (T), ondansetron (O), and granisetron (G) between 25 and 120 hours following chemotherapy administration. CR: complete response; MR: major response; mR: minor response; Fa: failure. There were no differences in response rates among the 3 5HT₃ receptor antagonists ($p = 0.527$).

CINV (Acute & Delayed): Turkish Cancer Patients (Tropisetron vs Granisetron)

Metode: Clinical trial komparatif (profilaksis CINV)
Populasi: 158 pasien kanker (kemo emetogenik sedang-tinggi)

Intervensi:

Tropisetron: 5 mg IV + dexamethasone 8 mg, lanjut 5 mg PO qd sampai 5 hari
Granisetron: 3 mg IV + dexamethasone 8 mg, lanjut 1 mg PO bid sampai 5 hari

Hasil:

CR acute emesis: 80,4% (tropisetron) vs 71,7% (granisetron) ($p = 0,877$)
CR delayed emesis (hari 2–5): 68,6% vs 76,1% ($p = 0,527$)
Bebas mual (hari 2–5): 37,3% vs 33,9% ($p = 0,949$)

Tropisetron dan granisetron hasilnya mirip juga!

Kemanan Tropisetron

Adverse Event	Tropisetron				Ondansetron
	Placebo (n=473)	0.5 mg (n=174)	2 mg (n=485)	5 mg (n= 176)	4 mg (n= 299)
bradycardia	50 (11%)	1 (1%)	50 (10%)	2 (1%)	47 (16%)
hypotension	31 (7%)	0 (0%)	40 (8%)	1 (1%)	27 (9%)
headache	15 (3%)	4 (2%)	19 (4%)	17 (10%)	8 (3%)
shivering	10 (2%)	1 (1%)	15 (3%)	0 (0%)	12 (4%)
hypertension	12 (3%)	2 (1%)	16 (3%)	1 (1%)	17 (6%)
skin reactions	5 (1%)	3 (2%)	7 (1%)	2 (1%)	6 (2%)
bronchospasm	6 (1%)	0 (0%)	5 (1%)	0 (0%)	2 (1%)
dizziness	5 (1%)	2 (1%)	2 (0%)	2 (1%)	1 (0%)

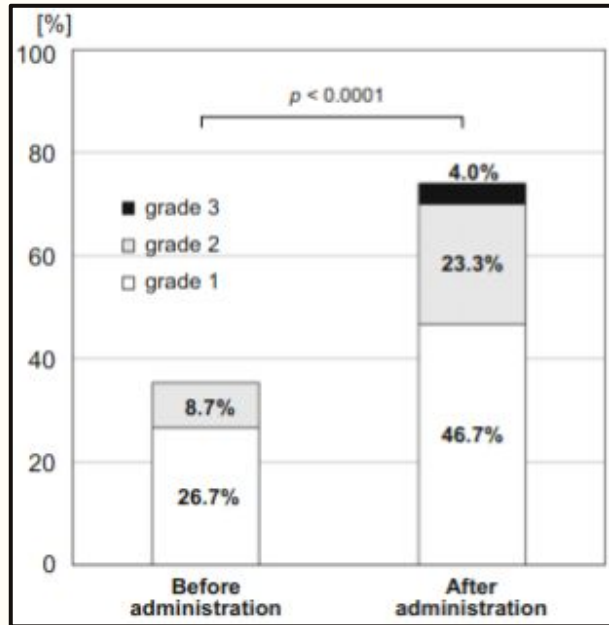
Tropisetron memiliki lebih sedikit efek samping dibandingkan Ondansetron

Kekurangan Granisetron

Efek samping granisetron: headache 14% dan konstipasi 3%
Ada laporan QT prolongation → perlu hati-hati pasien aritmia, gangguan elektrolit

Table 1 Principal Adverse Reactions in Clinical Trials — Single-Day Chemotherapy		
Percent of Patients With Reaction		
	Granisetron Hydro- chloride Injection 40 mcg/kg (n=1268)	Comparator ^a (n=422)
Headache	14%	6%
Constipation	3%	3%
^a Metoclopramide/dexamethasone and phenothiazines/dexamethasone.		
5.2 Cardiovascular Events		
An adequate QT assessment has not been conducted, but QT prolongation has been reported with granisetron hydrochloride. Therefore, granisetron hydrochloride should be used with caution in patients with pre-existing arrhythmias or cardiac conduction disorders, as this might lead to clinical consequences. Patients with cardiac disease, on cardio-toxic chemotherapy, with concomitant electrolyte abnormalities and/or on concomitant medications that prolong the QT interval are particularly at risk.		

Kekurangan Palonosetron



Palonosetron ada perhatian khusus pada konstipasi
Studi observasional retrospektif 150 pasien chemotherapy yang menerima palonosetron IV 0,75 mg + kemoterapi lini pertama
→ kejadian konstipasi meningkat dari 35,4% sebelum terapi menjadi 74,0% setelah terapi
4,0% mengalami konstipasi grade 3 yang memerlukan manual evacuation

Take Home Messages

Mual dan muntah adalah masalah klinis yang penting, karena dapat mengganggu pemulihan, intake oral, kenyamanan pasien, dan pada kondisi tertentu meningkatkan risiko komplikasi.

Jalur 5-HT₃ merupakan salah satu jalur kunci pada PONV, CINV, dan RINV, terutama melalui aktivasi serotonin di usus/serabut vagus menuju pusat muntah di otak.

Tropisetron bekerja sebagai antagonis reseptor 5-HT₃, sehingga menghambat transmisi sinyal emesis dan menurunkan kejadian mual-muntah.

Dari guideline, antagonis 5-HT₃ tetap menjadi pilar utama antiemetik, baik untuk profilaksis PONV maupun sebagai bagian regimen pada CINV dan RINV.

Jurnal jurnal klinis menunjukkan tropisetron efektif menurunkan PONV/CINV dibanding placebo atau terapi lama, dan efeknya dapat lebih baik bila dikombinasikan dengan dexamethasone.

Posisi tropisetron saat ini kuat untuk kontrol fase akut dan profilaksis perioperatif

Jadi tropisetron adalah opsi antiemetik yang evidence-based, guideline-supported, dan relevan untuk membantu kontrol mual-muntah pada setting PONV, CINV, dan RINV.